

a Form a local target

Three soft-SFA forces

SLOW
 $\omega_t(z_t - p_t)$

NONCOLLAPSE
 $-\frac{\lambda_v}{\lambda_s} g \odot (z_t - m)$

DECORRELATE
 $+\frac{\lambda_c}{2\lambda_s} A(z_t - m)$

$$r_t = r_t^{\text{slow}} + r_t^{\text{var}} + r_t^{\text{cov}}$$

$$\hat{z}_t = \text{sg}(z_t - r_t)$$

b Define the neuron state

Residual, then same-layer settling

activation residual

$$z_t - \hat{z}_t$$

$$\downarrow J_{\phi,t}^{\top}$$

base neuron state

$$b_t = J_{\phi,t}^{\top} (z_t - \hat{z}_t)$$

same-layer inhibition

$$b_t \text{ --- } S_t$$

$$(I + \kappa M)S_t \approx b_t$$

c Update local synapses

The settled state supplies both rules

$S_{t,j}$ branches into:

- FEEDFORWARD GRADIENT**
 $\nabla_{W_{ji}} \tilde{L}_a \propto S_{t,j} x_{t,i}$
- LATERAL CHANGE**
 $\Delta L_{jk}^{\text{lat}} \propto -S_{t,j} S_{t,k}$

Diagram showing $x_{t,i}$ connected to $S_{t,j}$ via W_{ji} .

LOCAL IN TIME

one detached predecessor p_t ; no graph through time

LOCAL IN SPACE

updates read endpoint states; no error crosses layers